Polyacids as Modulators for the Synthesis of UiO-66

Kyle C. Bentz, Sergio Ayala Jr., Mark Kalaj, and Seth M. Cohen*

Department of Chemistry & Biochemistry, University of California, San Diego, La Jolla, California, 92093, USA

Abstract: Poly(acrylic acid) (PAA) and poly(vinylbenzoic acid) (PBA) were synthesized via reversible-addition fragmentation chain transfer (RAFT) polymerization and used as modulators for the synthesis of the metal-organic framework (MOF) UiO-66 (UiO = University of Oslo). While typical syntheses of UiO-66 require large excesses of acid modulators, such as acetic acid or benzoic acid, to achieve controlled particle size and morphology of the resulting MOF particles, the use of polymerized acids allows for narrow particle size distributions at sub-stoichiometric quantities of modulator. We show using scanning electron microscopy (SEM) and dynamic light scattering (DLS) techniques that polyacids can act as alternative modulators for the growth of UiO-66.

Introduction

Metal-organic frameworks (MOFs) are porous network solids composed of organic multitopic ligands linked via metal clusters known as secondary building units (SBUs). Zr(IV)-based MOFs like UiO-66 (composed of $Zr_6(\mu_3\text{-}O)_4(\mu_3\text{-}OH)_4$ SBUs and 1,4-benzene dicarboxylic acid (H_2bdc)) have become a highly studied class of MOFs due to their high stability and chemical tunability. The solvothermal synthesis of UiO-66 is often accompanied by monotopic ligands known as modulators, typically acetic acid (AA) or benzoic acid (BA).[1] Modulators act to slow the growth of the MOF particles, resulting in narrower particle size distributions and controlled morphologies.[2] However, in order to achieve effective modulation of MOF growth, a large excess of the monofunctional acid is required. Furthermore, modulators are only effective over a narrow range of concentrations, that is, undermodulation can result in uncontrolled particle growth and overmodulation can preclude the formation of MOF entirely.

While there are several examples of polymers containing coordinating or structure directing blocks,[3] and even ligand-containing blocks,[4] the use of polymers explicitly as modulators is largely unreported. An early example of the use of polymers to control the growth of coordination polymers (CPs) was reported by Kitagawa, where the sodium salt of poly(vinylsulfonic acid) was used as an additive to control size in a Cu-based coordination polymer.[5] When the additive was present, the resulting CPs formed much larger crystals than when no additive was present due to the slowing of nucleation events. Polymers have also been used to template the growth of MOFs into various morphologies or confined geometries. Eddaoudi and coworkers used polyvinylpyrrolidone (PVP) in combination with small molecule quaternary ammonium compounds to provide excellent control of crystal size morphology in In(III)-based square-octahedral MOFs.[6] PVP was also used to selectively control the sizes of the [100] and [111] facets in cubic Zn(II)-based frameworks.[7] Triblock polymer surfactants have been used to provide both micro- and mesoporous MOFs by directing framework formation around the nanostructures formed from the block polymer assemblies.[8] There are several examples of the use of block polymers containing coordinating blocks to create higher order assemblies in MOF systems. For example, Cheetham and coworkers used poly(styrene-*b*-(4-vinyl pyridine)) to create hierarchical superstructures in ZIF-8,[9] and Schmidt and coworkers used poly((ethylene oxide)-*b*-(methacrylic acid)) to create large hexagonal mesocrystals in both Zn(II) and Cu(II)-based MOFs.[10] Coordinating block polymers have also been shown to act as modulators for the synthesis of metal-organic polyhedral (MOPs), where polymers containing poly(isophthalic acid) blocks effectively controlled the crystal habit in a variety of $Cu_{24}L_{24}$ cages.[11]

In addition to the benefit of eliminating large quantities of waste generated by the use of small molecular modulators in stoichiometric excess,[12] herein it is shown that polyacid modulators provide control of MOF growth over a wider range of concentrations when compared to simple molecular acids. In particular poly(acrylic acid) (PAA) was shown to provide MOF particles at sub-300 nm sizes with very narrow particle size distributions. This is consistent with the work of Lotsch and coworkers, who used

commercially available PAA to achieve small particles of HKUST-1 and IRMOF-3;[13] however, a systematic study of the effect of acid concentration was not carried out. While there have been studies that have elucidated the effect of concentration of small molecule modulators for UiO-66,[14] no such study with polymer modulators has been performed to date. Herein, we report the use of PAA and poly(vinylbenzoic acid) (PBA) as modulators for the growth of UiO-66 over a very wide range of concentrations, and find that both polymers act as modulators at low concentrations to control both the size and size distribution of UiO-66 nanoparticles.

Results and Discussion

The concept of using polyacids as a modulator for MOF growth stems from the hypothesis that while the total concentration of acid modulator with respect to organic linker may be sub-stoichiometric, the local concentration of acid still remains high (Figure 1). In contrast, the small molecule acid modulator always exists in the condition where the local concentration is equal to the total concentration. In other words, while 50 equivalents of small molecule acid are evenly distributed throughout the reaction medium, a polymer with a degree of polymerization (DP) of 50 contains 50 equivalents of acid on a single chain and therefore should behave analogously to a much higher concentration of small molecule acid. We therefore speculated that the use of polymer modulators would allow for controlled MOF particle growth at sub-stoichiometric quantities. Two different polyacids were used in this study, PAA and PBA, which are structurally analogous to AA and BA that are commonly used small molecular modulators for the synthesis of UiO-66. Reversible-addition fragmentation chain transfer (RAFT) polymerization was used to synthesize the PAA and PBA polymers, as RAFT allows for polymers of predetermined molecular weights and affords polymer with very narrow molecular weight distributions. ^{1}H NMR confirmed that DPs of 51 and 19 were achieved and gel permeation chromatography revealed narrow dispersities of $Đ$ = 1.04 and 1.10, for PAA and PBA, respectively (Figure S1, S2).

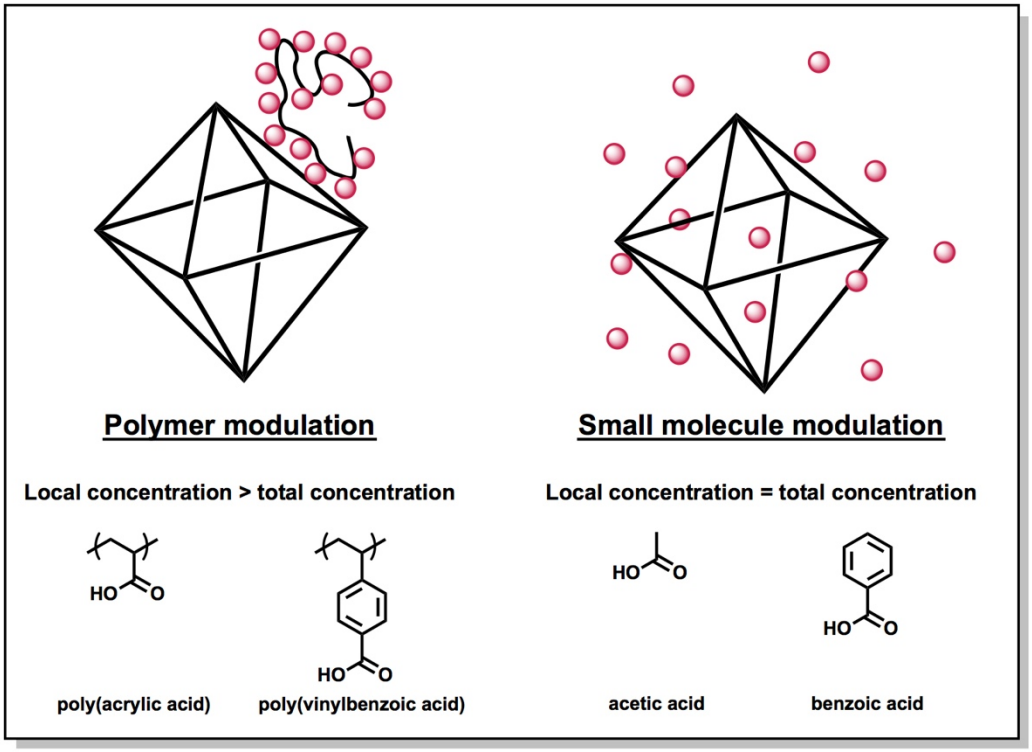


Figure 1. Schematic of polyacid modulation on the growth of MOF particles. Because the polyacids contain chains of covalently linked acids, the local concentration is greater than the total solution concentration of acid.

For a variety of applications in which MOFs are envisioned, control over particle size and size distribution is critical. Therefore, monotopic acid modulators are often introduced during synthesis to control particle nucleation and slow growth, leading to particles with narrow size distributions. In this study, particle sizes were characterized by both dynamic light scattering (DLS) and scanning electron microscopy (SEM). While SEM can provide useful information regarding particle morphology, accurate determination of particle size can be challenging, especially for samples with large size distributions. However, in the use of DLS provides an unambiguously, bias-free quantification of particle sizes and distributions. Particle sizes and distributions were calculated using a cumulants analysis of the second-order autocorrelation function $G^2(\tau)$ (for further details see ESI).

When AA was used as the modulator for UiO-66 synthesis, only a very narrow range of acid concentrations provided effective modulation of the MOF particles. Indeed, only when 30 equivalents of AA were used was the average particle size <300 nm as measured by both DLS and SEM (Figure 2). When >30 equivalents of AA was used, the resulting MOF particles grew markedly in size and the size distribution widened considerably. Addition of >300 equivalents of AA in the UiO-66 synthesis resulted in over modulation and no isolable solids were obtained. In contrast, the use of PAA allowed for a narrow distribution of particles sizes, as measured by both DLS and SEM, over a very wide range of concentrations (Figure 2). The use of PAA as the modulator for UiO-66 synthesis gave particles with diameters less than 400 nm as measured by DLS over the entire concentration range spanning greater than four orders of magnitude. Measurements of particle size using SEM images corroborate these results, indicating a very tight distribution of particle sizes when PAA is used as the modulator, giving highly controlled particles over the entire range of equivalents tested. While SEM shows a tighter distribution of particle size for AA modulated UiO-66, this can only be achieved in a very narrow range of modulator concentration (Figure 2). In addition to controlling the growth of UiO-66, the use of PAA also endows the resulting MOF particles with superior colloidal stability. Particles synthesized using AA sediment from solution quite rapidly (<1 day), but particles made under identical conditions but with PAA modulator remain exceptionally well dispersed after standing for one week (Figure S5). We hypothesize that this additional colloidal stability is due to coordinated polymer that remains exposed on the surface which provides stabilization and prevents aggregation events.

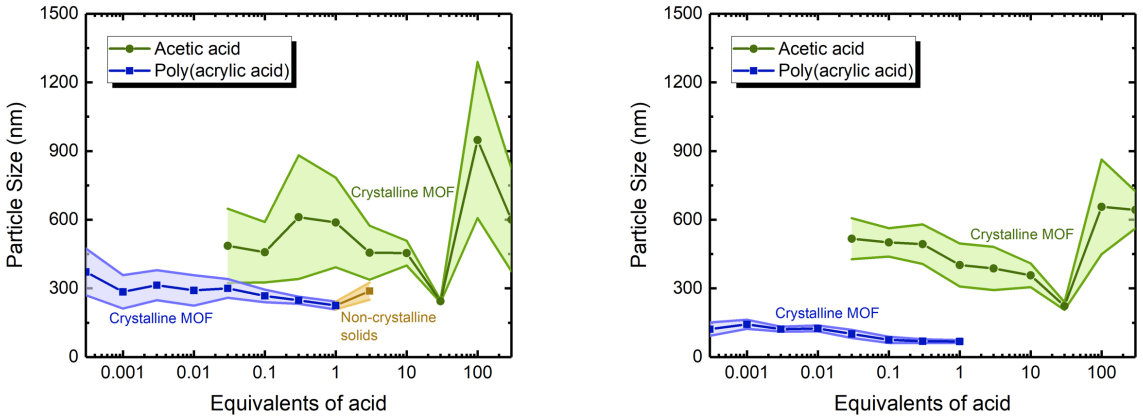


Figure 2. Particle size as a function of modulator content as determined by dynamic light scattering (DLS, *left*), and scanning electron microscopy (SEM, *right*). Shaded boundaries represent size distributions as determined by PDI (DLS) and standard deviation of particle size (SEM).

Across the entire range of acid equivalents tested with AA, the crystallinity of the particles was very high, as determined by powder X-ray diffraction patterns (PXRD, Figure 3). While low concentrations of small molecular modulator gave materials with uncontrolled particle size and broad size distributions, crystallinity remained high throughout the concentration range. However, when PAA was used as the

modulator for UiO-66 growth, crystalline particles were not produced until the concentration is reduced to stoichiometric equivalence with respect to H_2bdc ligand concentration. This observation supports the hypothesis that polymer acids have very high effective local concentration. In contrast to small molecule AA, which required >300 equivalents to achieve overmodulation and preclude particle growth, only three equivalents of PAA resulted in overmodulation, highlighting the efficiency of the polymer acid. Interesting to note as well, is that while overmodulation with AA causes no solids to form during synthesis, when UiO-66 growth is overmodulated by PAA non-crystalline solids are formed, likely composed of metal-coordinated crosslinked polymers. These metal-coordinated polymers may have interesting potential as amorphous MOFs or metallogels under the proper synthetic conditions.[15]

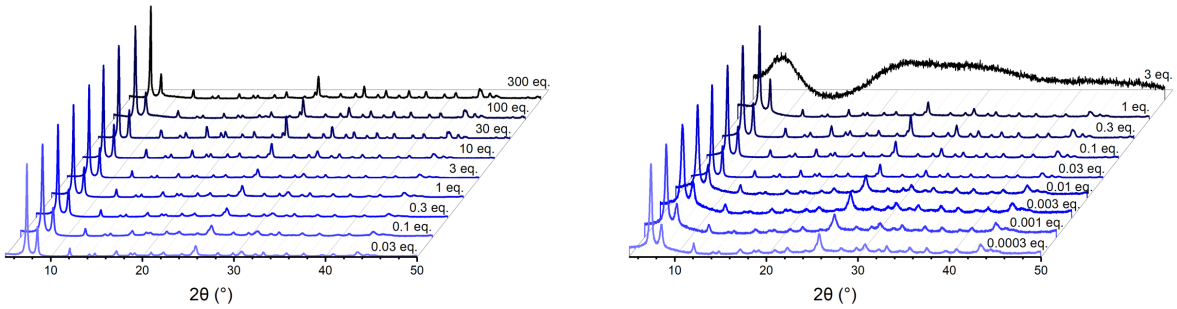


Figure 3. Powder X-ray diffraction patterns for UiO-66 synthesized at various equivalents of AA (*left*) and PAA (*right*).

Crystal morphology is largely unaffected by the use of PAA modulator at low concentrations (Figure 4). While particle sizes are smaller and size distributions are narrower when PAA was used as the modulator for UiO-66, highly faceted or regular geometries were not obtained. It has been previously shown that the use of doubly hydrophilic block polymers provides morphology control in Zn(II) and Cu(II)-based MOFs.[10] In this system, poly(methacrylic acid) acted as the coordinating block to slow the crystal growth, while a poly(ethylene glycol) block worked to stabilize the growing MOFs and provide assembly of mesocrystals into large, micron-sized hexagonal crystals. Perhaps the use of block polymers for modulation and morphology control in the growth of UiO-66 will result in similar particle morphology control, but that is outside the scope of this study.

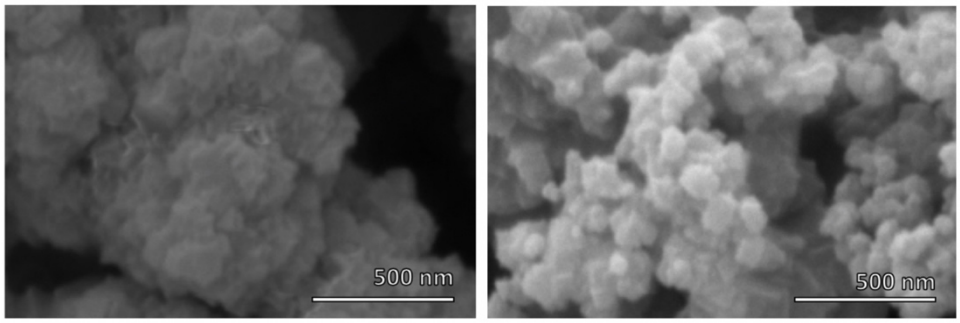


Figure 4. Scanning electron microscopy images of UiO-66 modulated by 0.03 equivalents of AA (*left*) and PAA (*right*). Particles modulated by PAA display lower levels of aggregation compared to those modulated by AA.

Additionally, BA and PBA were used to modulate UiO-66 (Figures S3, S4). When BA was used as a modulator, 10 equivalents provided particles with both the smallest diameters and the narrowest size distributions. Crystalline particles with narrow size distributions were produced when 30 equivalents of

BA were used, but at equivalents greater than 30, no particles were produced as a consequence of overmodulation. At <3 equivalents, modulation by BA becomes increasingly less controlled with very large particle sizes and size distributions. Modulation by PBA provided improved levels of control compared to BA, with very narrow size distributions afforded by equivalents of acid as low as 0.03. Below these levels, particle size and size distributions increased markedly. Interestingly, overmodulation with PBA occurred at >0.3 equivalents of PBA, demonstrating the remarkable modulation efficiency of this polymer. As in the case of PAA, overmodulation by PBA resulted in non-crystalline solids.

Lastly, BET surface area measurements were performed on select samples to access how modulator type and concentration affects the porosity of the resulting UiO-66 particles (Figures S6-S13, Table S1). Particles modulated with PAA showed a lower surface area compared to particles synthesized with 30 equivalents of AA, but were comparable to particles modulated with 1 equivalent of AA. In contrast, particles synthesized with PBA retained high surfaces areas, comparable to materials prepared with BA. When BA was used at 10 and 0.3 equivalents, surfaces areas of 1,046 ± 194 and 1,018 ± 86 were achieved, respectively. Similaraly, when PBAwas used at 0.3 and 0.03 equivalents, surface areas of 946 ± 13 and 1,010 ± 25 were achieved, respectively, indicating that PBA is an effective modulator for UiO-66 even at substoichiometric concentrations.

Conclusions

In conclusion, we have shown that polymer acids are useful modulators for the synthesis of UiO-66, providing narrow particle size distributions over a wide range of concentrations, and at equivalencies orders of magnitude lower than their small molecule analogues. Average particle sizes and size distributions were determined by both SEM and DLS techniques. In addition to providing efficient levels of modulation, synthesis of UiO-66 using PAA as a modulator resulted in particles with high colloidal stability, remaining well suspended in solution after more than one week sitting undisturbed.

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Conflicts of Interest

The authors declare no conflicts of interest.

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